Aerospace Manufacturing Core
Study Guide

"AMCSG-101"

Introduction to Aerospace Manufacturing

Airplanes & Flight

Aircraft, Airplanes, and Assemblies

Text from Google AI

A Brief History of Aviation

- **1783**: First passenger flight (hot air balloon)
- **1903**: First airplane flight (Wright brothers)
- **1916**: Boeing Model 1 (B&W Seaplane)
- **1957**: First Commercial Jet (707)
- **1969**: First Commercial Jumbo Jet (747)
- **1994**: First Commercial CAD Jet (777)
- **2007**: First Commercial Composite Jet (787)

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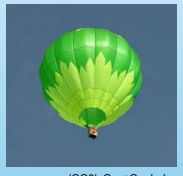
- Google “First hot air balloon" = (CC4) https://commons.wikimedia.org/wiki/File:Men_and_women_with_telescopes,_gyroscopes,_a_hot-air_balloon_Wellcome_V0040858.jpg
- Google “Wright Brothers first flight” = (PD) https://www.rawpixel.com/image/8718008/photo-image-vintage-airplane-public-domain
- Google “Boeing Model 1 B&W Seaplane” = (PD) https://en.wikipedia.org/wiki/Boeing_Model_1
- Google “Boeing 707” = (PD) https://jenikirbyhistory.getarchive.net/media/boeing-707-131b-trans-world-airlines-twa-jp5946969-6d93d6
- Google “Boeing 747” = (CC2) https://commons.wikimedia.org/wiki/File:Boeing_747-400_Dreamliner_livery.jpg
- Google “Boeing 777” = (CC2) https://commons.wikimedia.org/wiki/File:Boeing_777_above_clouds,_crop.jpg
- Google “Boeing 787” = (CC2) https://commons.wikimedia.org/wiki/File:Boeing_787_first_flight.jpg

Aircraft vs. Airplane?

- **Aircraft** = any vehicle that is capable of flight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- **Airplane** = any winged vehicle, heavier than air, driven by engine

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- Google "Hot air balloon" = (CC0) - https://pixnio.com/sport/hot-air-ballooning/hot-air-balloon-greeley
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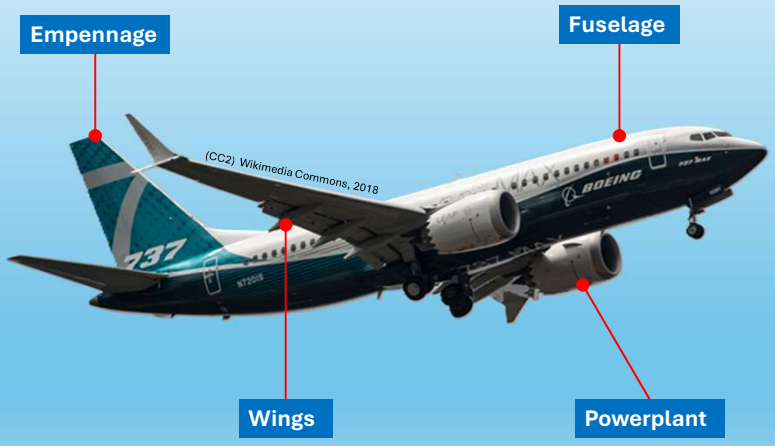
The Four Primary Assemblies



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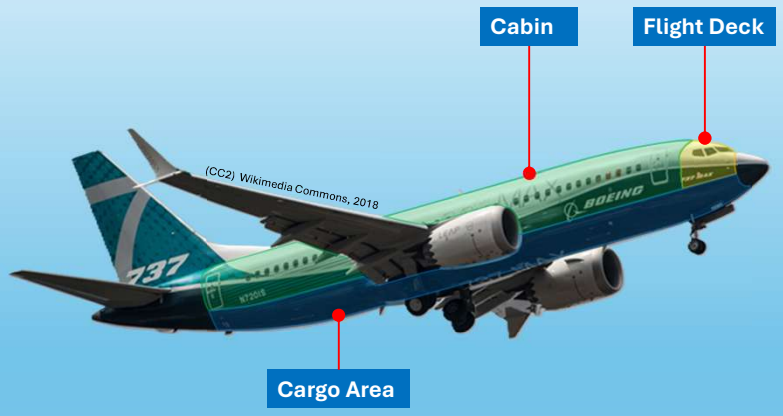
The Fuselage



Flight Deck (cockpit) is where flight crew sites. Contains flight control systems: avionics, communications, and navigational control.

Cabin is where the passengers sit, seats exist, etc.

Cargo Area is where luggage is stowed

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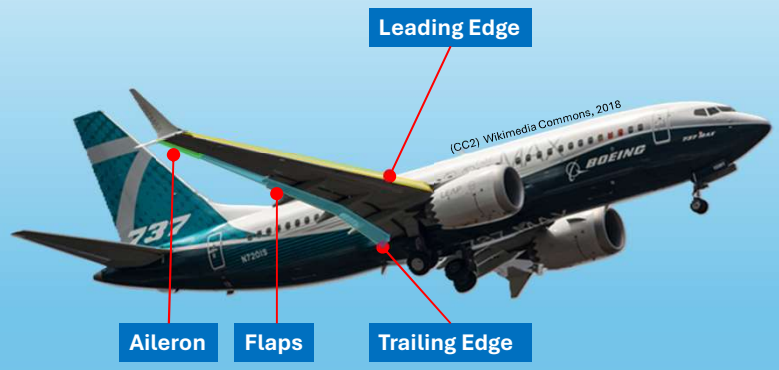
The Wings



Leading Edge is front edge of wing that contacts air first during flight, separating it above and below for lift

Trailing Edge is rear edge of wing where air flow comes back together with the top flow

Ailerons are small flaps at the trailing edge of the wing, far out near the tip; used to roll a plane by moving the left and right ailerons in opposing directions (up and down). Control pivoting around the longitudinal axis that runs nose to tailtip.

Flaps are what you think of as the trailing edge hinged control surfaces that make the wing larger or smaller, deeper or shallower to reduce stall speed for landing, reduce drag for flight, etc.

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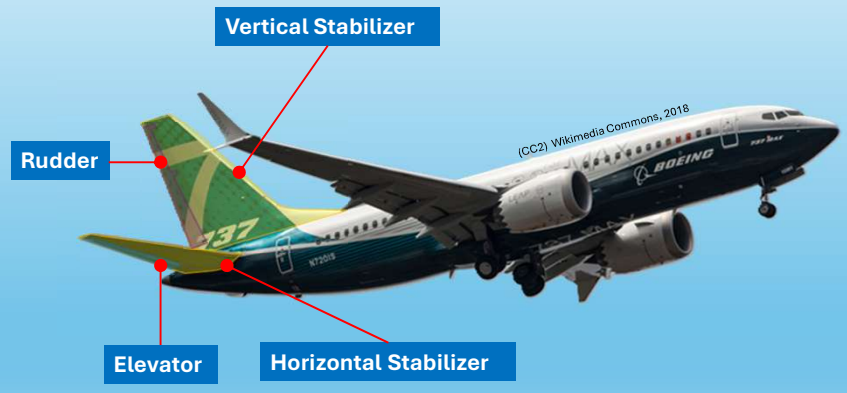
The Empennage



Vertical Stabilizer is the tail (the 90% that is stationary)

Rudder is the hinged part of the tail that controls left to right turning about the vertical axis

Horizontal Stabilizer is the small wing attached below the tail

Elevators are the small hinged flaps on the horizontal stabilizer that allow up or down pitch of the airplane along the lateral axis (wingtip to wingtip)

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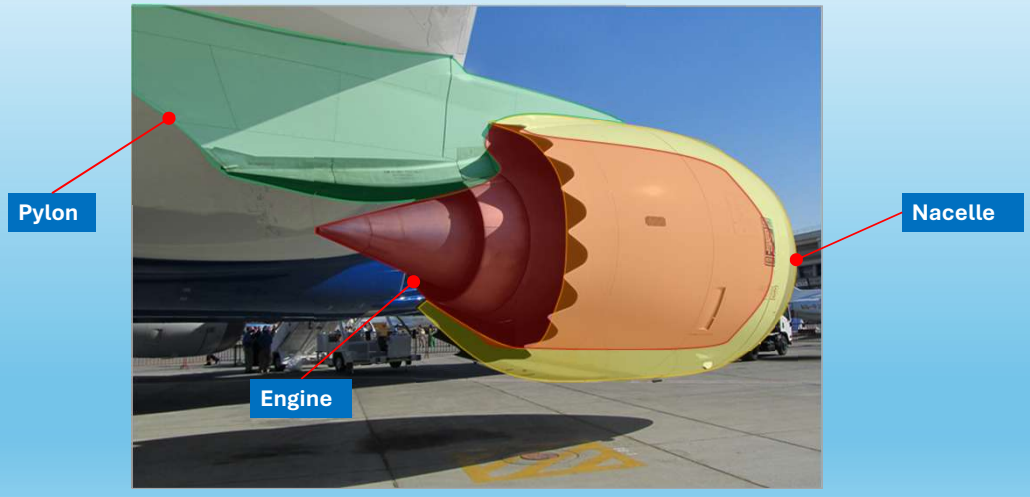
The Powerplant



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- **Engine** for Thrust
- **Nacelle** to cover it for aerodynamic flow (less drag)
- **Pylon** to fasten it all to the wing

Image Credits:

- Google “boeing jet engine pylon nacelle picture“ = (CC2) - https://commons.wikimedia.org/wiki/File:Rolls-Royce_Trent_1000_%287976551649%29.jpg

Principles of Flight

Text from ChatGPT

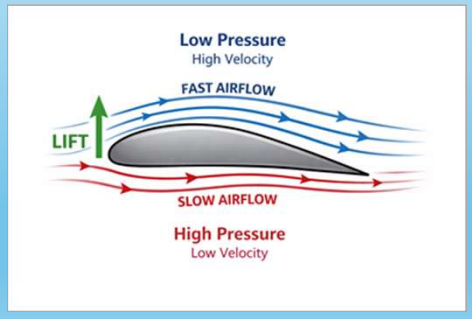
The “Four Forces of Flight”

ChatGPT / Matt Pierce, 2026

- **Gravity** pulls downward (-)
- **Thrust** / Engine pulls forward (+)
- **Drag** resists thrust, backward (-)
- **Lift** of airfoil pulls upward (+)

Airfoil works by…

- “Bernoulli principle”
- Longer distance around top of wing
- Creates faster airflow / lower pressure above
- Deflects air downward causing upward force



ChatGPT / Matt Pierce, 2026

Image Credits:

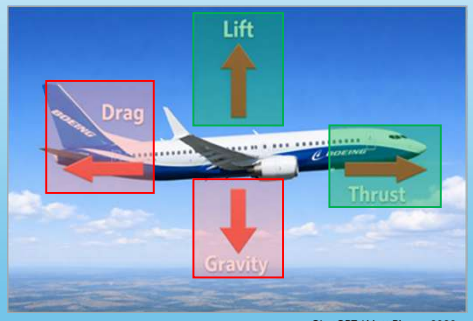
- 4 Forces = ChatGPT = me - “make a picture with boeing jet having red arrows depicting thrust, drag, lift, and gravity” >> “show the image from a side view”
- Airfoil diagram = ChatGPT = me - “create diagram showing airfoil, lift, low pressure high velocity above, high pressure low velocity below”

Text from ChatGPT

The “Power-to-Weight Problem”

“**Thrust** must be great enough to overcome **Drag**...and induce **Lift** too.”

- Reduce Negative Forces:
  - Reduce **Gravity**:
    - Lighter, stronger materials
    - More efficient engines = less fuel/weight
  - Reduce **Drag:**
    - Better aerodynamic design
- Increase Positive Forces:
  - Increase **Thrust:**
    - More powerful engines
  - Increase **Lift:**
    - Better aerodynamic design (e.g.: winglets)



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Q&A

Practice Q&A's (Q1-Q9)

Q1: What force pulls an airplane down?

A: Gravity

Q2: What force pushes an airplane up?

A: Lift

Q3: What force pushes an airplane forward?

A: Thrust

Q4: What force slows an airplane down?

A: Drag

Q5: What is the shape of an airplane wing called?

A: Airfoil

Q6: What does a barometer measure?

A: Atmospheric pressure

Q7: What force does a wing or airfoil create?

A: Lift

Q8: What is the aerodynamic cover over a powerplant called?

A: Nacelle

Q9: What part of the airplane mounts the powerplant (engine) to the wing?

A: Pylon

Practice Q&A's (Q10-Q17)

Q10: What part of the empennage controls yaw (left/right pointing)

A: Rudder

Q11: What part of the empennage controls pitch (up/down pointing)

A: Elevator

Q12: What part of the wing controls roll?

A: Aileron

Q13: What is the official name of the tail of an airplane?

A: Empennage

Q14: What part of the airplane makes thrust?

A: Powerplant

Q15: What is the difference between an "aircraft" and an "airplane"?

A: Airplane must have wings and a powerplant.

Q16: What three areas does the fuselage breakdown into?

A: Fight deck, cabin, and cargo areas

Q17: What part of the wings produce up and down forces?

A: Flaps

Q: What are the 4 primary assemblies of an airplane?

A: Fuselage, empennage, wings, and powerplants

Thank You

Matt Pierce - mpierce710006@gmail.com - linkedin.com/in/mpiercead576/

https://github.com/DataResearchLabs/study_guide_intro_aerospace_mfg